

Final Review, MAT 2540 Professor Chiu

- This review consists of 13 sets of questions.
- You have 100 minutes to complete this review.
- Show all work and justify your answers.
- Wishing you success.

1. Let $P(n)$ be the statement that a postage of n cents can be formed using just 5-cent stamps and 7-cent stamps.

Use **strong induction** to prove that $P(n)$ is true for all integers $n \geq 24$.

2. Give a recursive definition of the sequence $\{a_n\}, n = 1, 2, 3, \dots$, if

a) $a_n = 4n - 2$. b) $a_n = n(n + 1)$. c) $a_n = 1 + (-1)^n$. d) $a_n = n^2$

3. Determine whether the given proposed definitions is a valid recursive definition of a function f from the set of nonnegative integers to the set of integers.

$$f(0) = 2, f(n) = f(n - 1) \text{ if } n \text{ is odd, } n \geq 1 \text{ and } f(n) = 2f(n - 2) \text{ if } n \text{ is even, } n \geq 2.$$

(a) If f is well defined, find a formula for $f(n)$ when n is a nonnegative integer.

(b) Prove that your formula is valid by induction.

4. (a) How many numbers must be selected from the set $\{1, 3, 5, 7, 9, 11, 13, 15\}$ to guarantee that at least one pair of these numbers add up to 16?

(b) Show that among any group of four (not necessarily consecutive) integers, there are two with the same remainder when divided by 3.

5. (a) Find a recurrence relation for the number of ways to completely cover a $2 \times n$ checkerboard with 1×2 and 2×2 dominoes with the initial conditions for the recurrence relation.

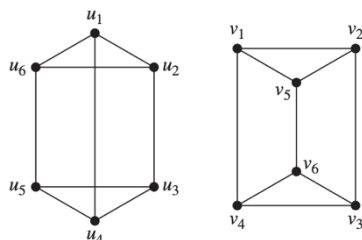
(b) Use mathematical induction to verify the formula.

6. Find the general form of the particular solution of the linear nonhomogeneous recurrence relation

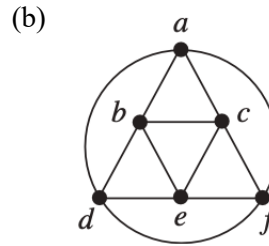
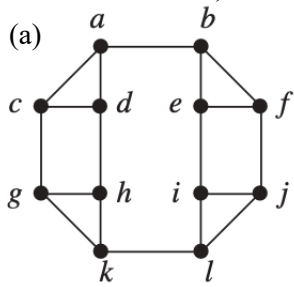
$$a_n = 4a_{n-1} - 4a_{n-2} + F(n) \text{ if (a) } F(n) = n^3 \quad \text{(b) } F(n) = n^2 4^n \quad \text{(c) } F(n) = n^4 2^n$$

7. Find $f(n)$ when $n = 3^k$, where f satisfies the recurrence relation $f(n) = 2f\left(\frac{n}{3}\right) + 4$ with $f(1) = 1$.

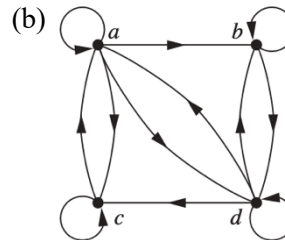
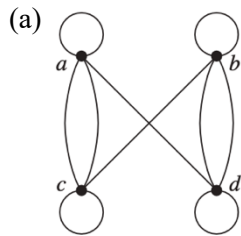
8. Given two graphs G and H. Find the number of vertices, edges, and degrees of vertices respectively and check if they are isomorphic. If yes, then find the isomorphism.



9. Find the vertex cut, the vertex connectivity, the edge cut, and the edge connectivity of each give graph.

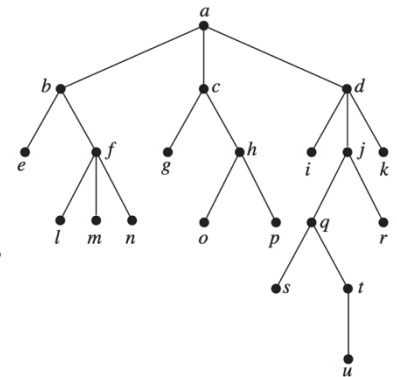


10. Find the degree, or in/out-degree of each vertex and the adjacency matrix of the given graphs with respect to the vertices listed in alphabetic order.



11. Answer these questions about the rooted tree illustrated.

- (a) Which vertex is the root?
- (b) Which vertices are internal?
- (c) Which vertices are leaves?
- (d) Which vertices are children of j ?
- (e) Which vertex is the parent of h ?
- (f) Which vertices are siblings of o ?
- (g) Which vertices are ancestors of m ?
- (h) Which vertices are descendants of b ?
- (i) Find the level of each vertex in the given rooted tree. What is the height of this tree?
- (j) Is the given rooted tree a full m -ary tree? Why or why not?



12. (a) Build a binary search tree for the words *banana*, *peach*, *apple*, *pear*, *coconut*, *mango*, and *papaya* using alphabetical order.

(b) Given the algorithm of Binary Search Tree. How many comparisons are needed to locate or to add each of these words in the search tree for (a), starting fresh each time?

(1) *kumquat*

(2) *pear*

ALGORITHM 1 Locating an Item in or Adding an Item to a Binary Search Tree.

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procedure insertion( $T$ : binary search tree,
 $x$ : item)
 $v := \text{root of } T$ 
{a vertex not present in  $T$  has the value null }
while  $v \neq \text{null}$  and  $\text{label}(v) \neq x$ 
    if  $x < \text{label}(v)$  then
        if left child of  $v \neq \text{null}$  then  $v := \text{left child of } v$ 
        else add new vertex as a left child of  $v$  and set  $v := \text{null}$ 
    else
        if right child of  $v \neq \text{null}$  then  $v := \text{right child of } v$ 
        else add new vertex as a right child of  $v$  and set  $v := \text{null}$ 
if root of  $T = \text{null}$  then add a vertex  $v$  to the tree and label it with  $x$ 
else if  $v$  is null or  $\text{label}(v) \neq x$  then label new vertex with  $x$  and let  $v$  be this new vertex
return  $v$  { $v$  = location of  $x$ }
    
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13. Given the coding scheme a: 001, b: 0001, e: 1, r: 0000, s: 0100, t: 011, x: 01010.

- (a) Show the given code is a prefix code.
- (b) Find the word represented by 01110100011.